

Plan for Microproject Mathematics

→ Topic → System performance bench-
-marking

↳ Using: Measures
of Tendency

Points

- What is Measures of Tendency.
- Tendency is the midpoint of a database
- How System performance bench-
-marking works.
- It works on define work loads.
- How tendency is related to
benchmarking.
- Both relatively doing measurements.
- What are the software and
hardware requirement for it

→ What type of language is used?

Python

→ Any extensions or derived
software?

→ Tendency: Since it is the avg of a database it can be determined by the use of mean

$$\text{mean} = \frac{\sum_{i=0}^n x_i}{n}$$



$$\text{Geometrical mean} = \sqrt[n]{\prod_{i=0}^n x_i}$$

$$\text{Harmonic mean} = \frac{n}{\sum_{i=0}^n \frac{1}{x_i}}$$

median ⇒ The midpoint of the database when the data are stored in ascending order.

mode: The highest occurrence data in the database.

Tendency  Benchmark



Storing the data
(performance) of
the system

Executing the
stored data and
predict system
performance

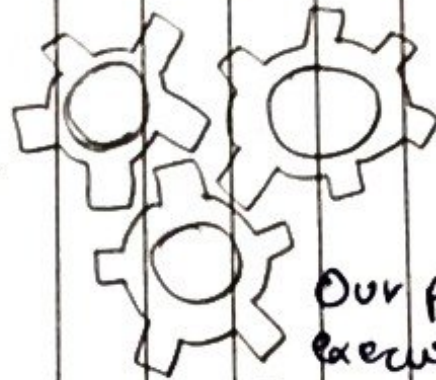
USELINK BETWEEN TND & BNM

Real time Benchmarking	Tendency Factor	what will it do?
1) Execution time	Normal mean	Actually reflects total elapsed time
2) Rate/Throughput	Harmonic mean	Correctly calculates and average correct rates eg: MB/s, FPS or IPC
3) Normalised Ratio	Geometric mean	Basically creates a baseline for running a system (also known as SPEC benchmark)
4) Response-latency	median	As by factor it checks the response of the benchmarking

Working in deep for 40/40



Getting information
from task manager



Our program
executing using
Tendency measures

Through
Pipelining

DASHBOARD

- Showing the real time Benchmarks

Operation testing